

Problem Definition

Misogyny online takes many forms, from obvious slurs to subtle stereotypes, sarcastic emojis, and memes. Current models struggle with implicit forms and are biased toward gendered terms.

Our goal: build a progressive pipeline that detects misogyny across text, emojis, and images, and understand where and why models fail.

We address three questions:

- Can models detect implicit misogyny, not just explicit slurs?
- Are models biased toward gendered words like "women" or "feminist"?
- Does adding emojis or images improve detection?

Related Works

- Devlin et al. (2019) / Liu et al. (2019) — BERT & RoBERTa: pretrained transformers for text classification, strong on explicit content but struggle with subtle forms.
- Kirk et al. (2022) — Hatemoji: standard models fail on emoji-based hate speech, but no analysis of gender-term bias.
- Kirk et al. (2023) — EDOS: fine-grained sexism labels, but models only evaluated on in-distribution data.
- Fersini et al. (2022) — MAMI: image+text beats text-only for memes, but no cross-domain evaluation.

→ Our work combines all three modalities, evaluates cross-domain generalization, and explicitly measures gender-term bias and implicit sexism detection.

Method

We build a progressive pipeline, each step adding a new modality or analysis layer on top of the previous one.

STEP 1	STEP 2	STEP 3
Can emojis help detect misogyny?	Can domain adaptation improve generalization?	Can visual features help for memes?
RoBERTa + emoji vs. RoBERTa	RoBERTa + emoji + OCR vs. RoBERTa + emoji	RoBERTa + CLIP vs. RoBERTa + emoji + OCR + CLIP vs. RoBERTa synthetic + CLIP vs. CLIP image only
Online Misogyny	Online Misogyny	MAMI

ANALYSIS		
Where and why do models fail?	Implicit vs. Explicit	EDOS
RoBERTa + emoji + OCR	Genre Bias	Online Misogyny
	Synthetic Benchmark	EDOS augmented vs EDOS

Key design choices:

- Emojis converted to text: 🙄 → :face_with_rolling_eyes: to preserve meaning without changing vocabulary
- Cross-domain evaluation: train on EDOS+Hatemoji, test on Online Misogyny for more honest generalization
- CLIP frozen: MAMI too small to reliably fine-tune a large-scale pretrained model.
- Concatenation fusion: simplicity and robustness over more complex cross-modal attention on limited data.

Data Sets

- **EDOS** (SemEval 2023). 20,000 Reddit/Gab posts — fine-grained sexism labels. Used for: training (Steps 1 & 2) + implicit/explicit analysis
- **Hatemoji**. 5,912 short statements with emoji-based hate speech. Used for: training emoji-aware RoBERTa (Step 2)
- **Online Misogyny** (EACL 2021). 6,356 Reddit posts (texts and images) — 8.1% misogynistic. Used for: cross-domain test set (Steps 2 & 3)
- **MAMI** (SemEval 2022). ~10,000 misogynistic memes (image + text). Used for: multimodal training and evaluation (Step 4)
- **Synthetic benchmark** (AI-generated) 300 controlled examples: 100 explicit, 100 implicit, 100 neutral/feminist + 50 emoji minimal pairs. Generated with fixed prompts and predefined schema (sexism type, category, emoji role, rationale)

Validation

Overall results across all models:

MODEL	F1	TEST SET
RoBERTa (no emoji)	0.719	Online Misogyny
RoBERTa + emoji	0.715	Online Misogyny
RoBERTa + emoji+ OCR	0.738	Online Misogyny
CLIP image only	0.789	MAMI
RoBERTa + CLIP	0.849	MAMI
RoBERTa synthetic + CLIP	0.849	MAMI
RoBERTa +emoji+ OCR + CLIP	0.855	MAMI

Key finding 1 — Implicit vs explicit sexism: - Explicit recall: 0.81 vs Implicit recall: 0.56 - Gap of 25 points, models rely on surface-level cues

Key finding 2 — Gender-term bias: - FP rate "female": 23.8% vs baseline 1.1% - Model flags neutral feminist discourse as misogynistic

Key finding 3 — Synthetic benchmark: - Implicit recall jumps from 0.06 to 0.82 with augmentation - But doesn't transfer to real EDOS data (0.556 → 0.490)

Key finding 4 — Multimodal: - Visual features improve meme detection consistently - Fine-tuned text encoder + CLIP achieves best F1 (0.855)

Limitations

- Implicit misogyny remains a blind spot: 25-point recall gap on real data, 61-point gap on synthetic benchmark
- Gender-term bias: "female" flagged 23.8% vs 1.1% baseline
- No emojis in Online Misogyny test set (0%) — emoji component partially unvalidated on real data
- No real-world dataset combines fine-grained implicit/explicit labels with emoji-rich content — hence our synthetic benchmark

Conclusion

- Transformer models handle explicit misogyny well but consistently miss implicit forms like stereotypes and backhanded compliments - Models develop a strong bias toward gendered terms, risking over-censorship of feminist discourse in practice
- Emojis have a smaller impact than expected on final predictions, though they can shift confidence in borderline cases
- Combining text and visual features significantly improves meme classification (F1 = 0.855 on MAMI)
- The core challenge is not data quantity but model understanding: current transformers rely on surface-level cues rather than grasping social and pragmatic context

→ Future work: context-aware models, emoji-rich evaluation datasets, better cross-domain benchmarks

References

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